plt_sim version 3

JLT, SAA

2025-06-21

History

- From plt_sim_v2.qmd 2025-06-14
- From plt_sim.R 2025-05-29

Simulation

Goal: Automate multiple simulations to demonstrate annual values of *run size* and spawning parameters (*arrival_timing, arrival_spread, exit_lag, exit_spread*) are recovered by fitting model spawner_m8.stan. All four spawning params as groups (multi-level regression) of annual values.

Simulation parameters

Assign distributions to parameters for runs and spawner behaviour. Set years and days for surveys. Params logged or not as per JLT code: spawners_m8.stan

```r
log_run_mu                <-  10  # log value for lognormal PDD,
log_run_sigma             <-   1  # log value for lognormal PDD,
arrival_timing_mu         <-  35  # normal
arrival_timing_sigma      <-   5
arrival_spread_mu         <-   5    # lognormal
arrival_spread_sigma      <-   1.35 # log value was 0.3
exit_lag_mu               <-  10    # lognormal
exit_lag_sigma            <-   1.35 # log value was 0.3
exit_spread_mu            <-   5    # lognormal
exit_spread_sigma         <-   1.35 # log value was 0.3
rnbinom_size              <-  10    # sampling error
```

```r
years_n                    <-  12    # 4x3 plot
days_n                     <- 100    # checked below
pct_surveyed               <-   0.9  # runif frac days sampled
# derived
days      <- 1:days_n
years     <- 1:years_n

# check range for days. pnorm(2.326348) = 99%
k99 = qnorm(0.99,0,1)
day_first=arrival_timing_mu -
    k99*(arrival_spread_mu+k99*arrival_spread_sigma)
day_last=arrival_timing_mu +
    k99*(arrival_spread_mu+k99*arrival_spread_sigma) +
    exit_lag_mu +
    k99*(exit_lag_sigma+exit_spread_mu+k99*exit_spread_sigma)
cat("Range of days with spawners conceivably present:", round(day_first),'to',round(day_last)
```

```
Range of days with spawners conceivably present: 16 to 86
```

Draws for annual values

Draw random samples for annual runs and spawning params. Spawners present each day is cumulative arrived minus cumulative exited based on those draws.

```
 run 95%CL: 3103 156367
    draws: 37130 7978 19312 7319 24139 28618 357118 10869 16095 29366 13469 15455

arrival_timing  95%CL: 25.2 44.8
    draws: 36.64 36.11 38.75 35.47 35.41 35.81 42.19 35.2 37.95 27.61 31.48 34.81

arrival_spread  95%CL: 2.78 9
    draws: 6.37 3.39 5.31 7 2.82 3.02 5.08 9.9 5.42 7.11 3.42 3.78

exit_lag  95%CL: 5.55 18.01
    draws: 5.92 9.55 6.47 13.11 17.2 9.54 13.69 12.13 17.57 10.78 6.73 15.23

exit_spread  95%CL: 2.78 9
    draws: 3.97 5.25 7.06 6 7.55 3.39 4.51 4.73 3.09 7.07 10.36 4.69
```

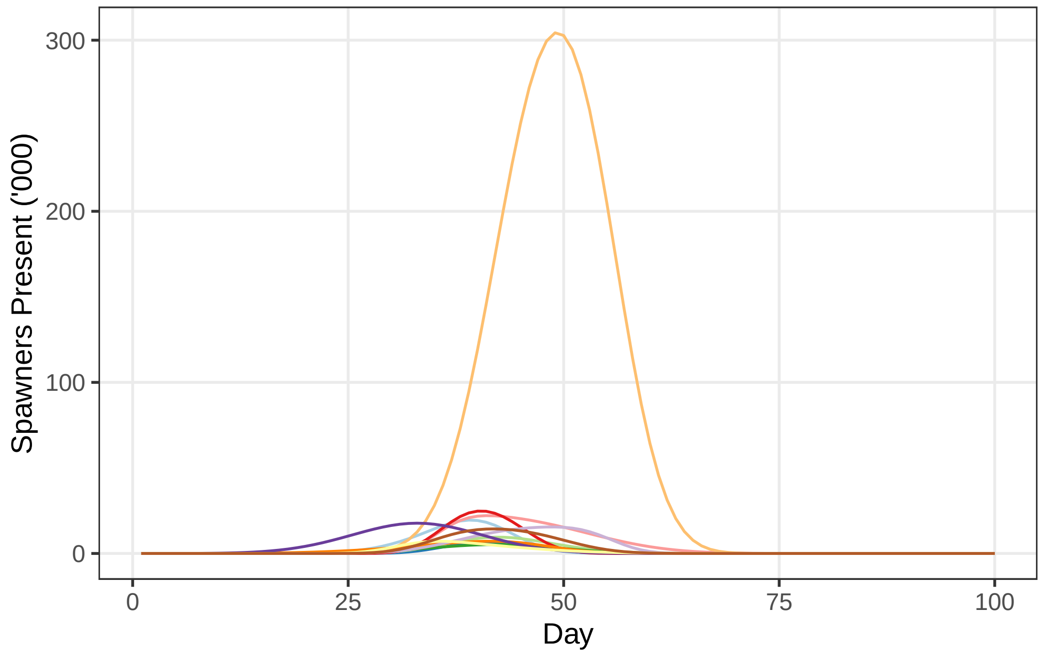


Figure 1: Time series of spawners present in each year.

Surveys

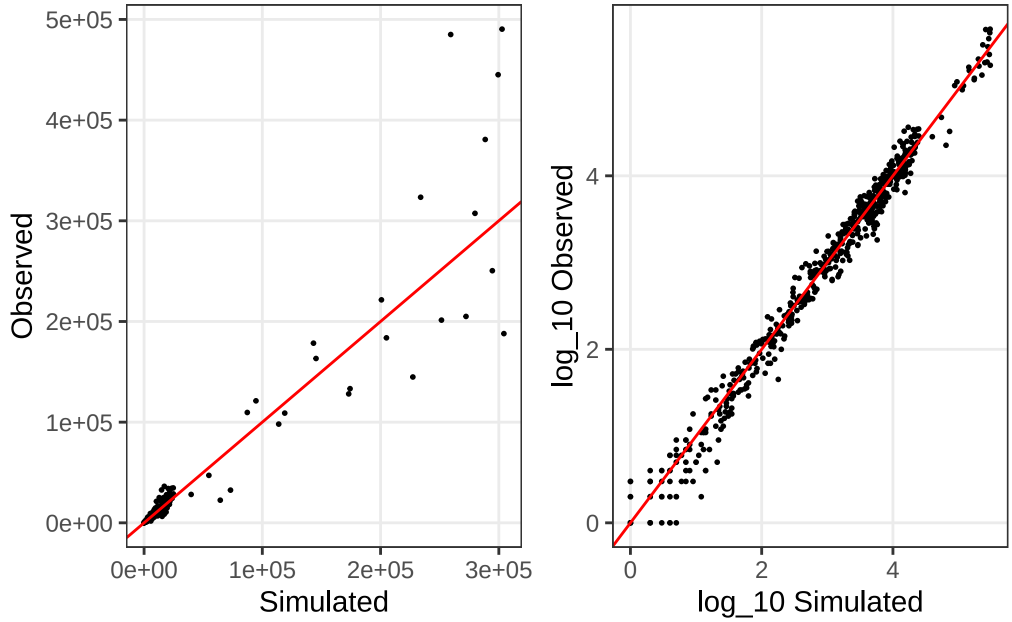
Define surveys as a random fraction of all days. Add negative binomial sampling error. Remove simulated values of zero and sampled observations of zero.

```
[1] max gap each year:
```

```
 [1] 1 2 3 1 2 1 2 1 1 1 2 1
```

Observed time series

Plotted as observed to show scatter near peak (mode) day and as logs to better show negative skew when arrival_spread is greater than exit_spread (typical), and positive skew when smaller.



(a) Sampling error is negative binomial. Note log(0) = 1.

Figure 2: Comparison of spawner observed to spawners present. Note log(0) = 1.

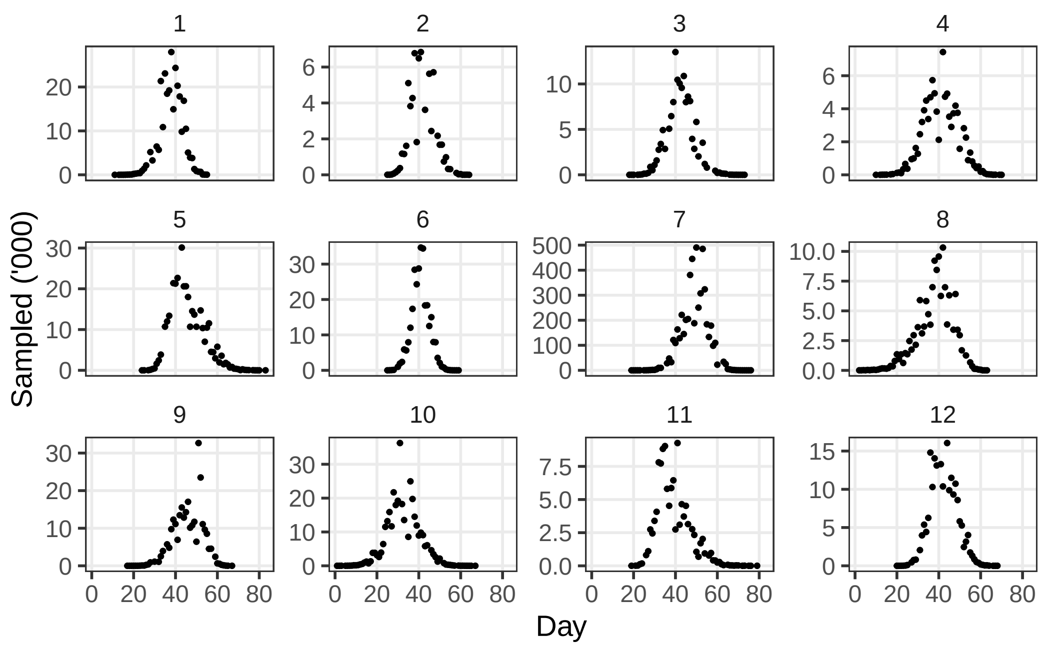
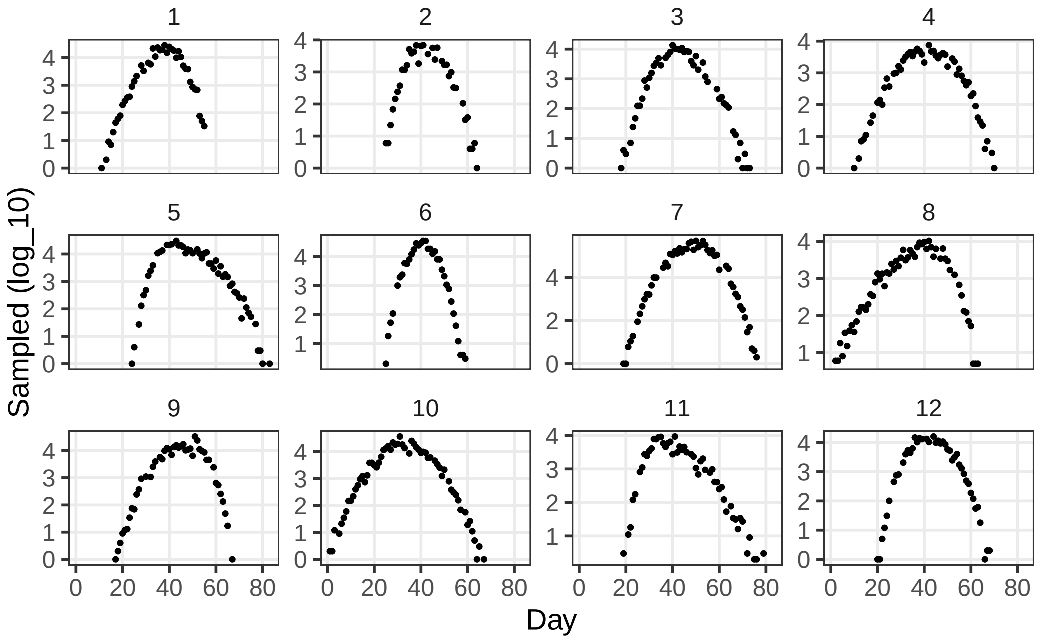


Figure 3: Annual time series of spawners observed.



(a) Reveals skewed distributions.

Figure 4: Annual time series, logged, of spawners observed.

Fit

Priors for Stan

Bayesian priors affect Markov Chain Monte Carlo (or related) sampling of the parameters hyperspace. The spawning parameters: *arrival_timing, arrival_spread, exit_lag, exit_spread* are treated as multilevel, groups of annual values. Priors fo a group define an initial distribution for the annual values: $exit\_lag = \Phi(\mu, \sigma)$ has group parameters *exit_lag_mu* and *exit_lag_sigma*. Annual values are sampled (fitted) from the group distribution, and group parameters are simultaneously fitted, thereby "sharing information" to improve the fit for each annual value. Note *log_runs* is not a group, the same prior applies to each year.

In what follows, the prior *exit_lag_sigma* uses the value for *spread_sigma*. The Stan code renames *count_dispersion* as *log(live_phi)* where *live_phi* is the size parameter for the negative binomial distribution.

Table 1: Priors for Stan sampling of parameters space.

prior	pdd	v1	v2
log_runs_mu	lnorm	10	1.0
arrival_mu	norm	40	5.0

Table 1: Priors for Stan sampling of parameters space.

prior	pdd	v1	v2
arrival_sigma	exp	0	10.0
spread_mu	norm	5	1.0
spread_sigma	exp	0	5.0
exit_lag_mu	norm	10	2.0
count_dispersion	norm	1	0.2

Data list for Stan

The priors are part of the Stan data list. Convergence of the HMC sampling chains is improved by providing initial values for *arrival_mu*. .

Stan sampling

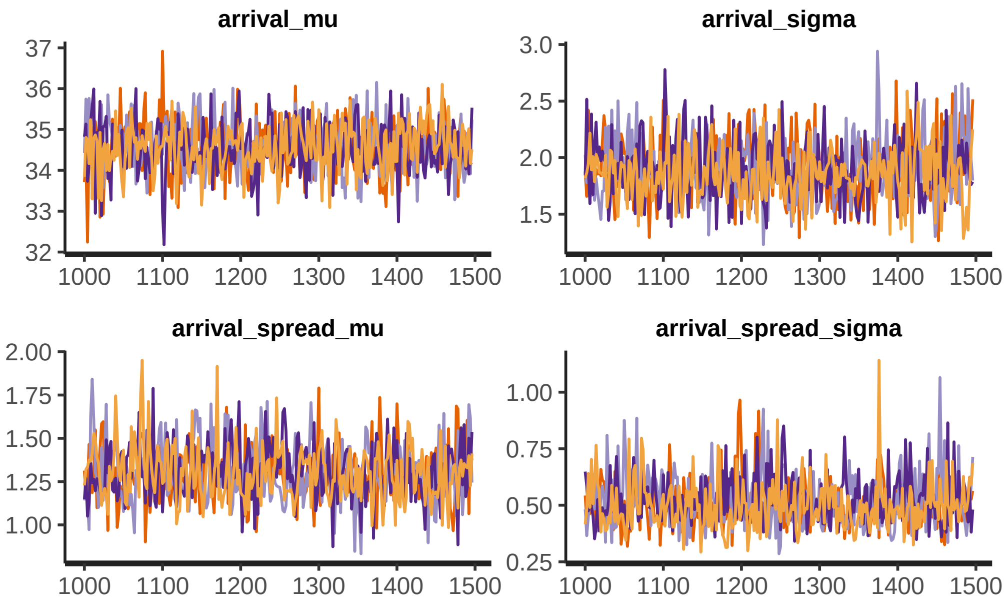
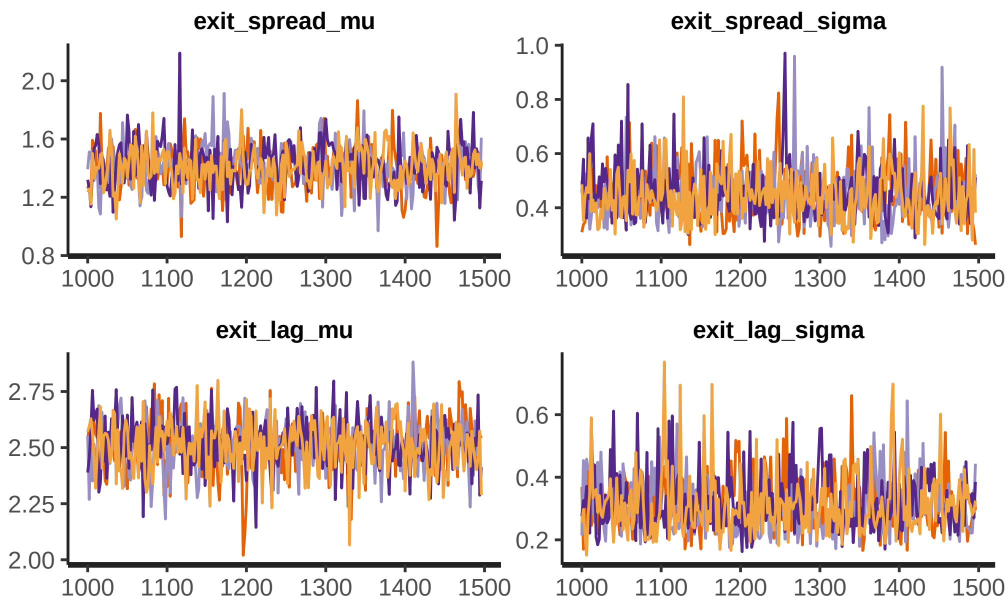
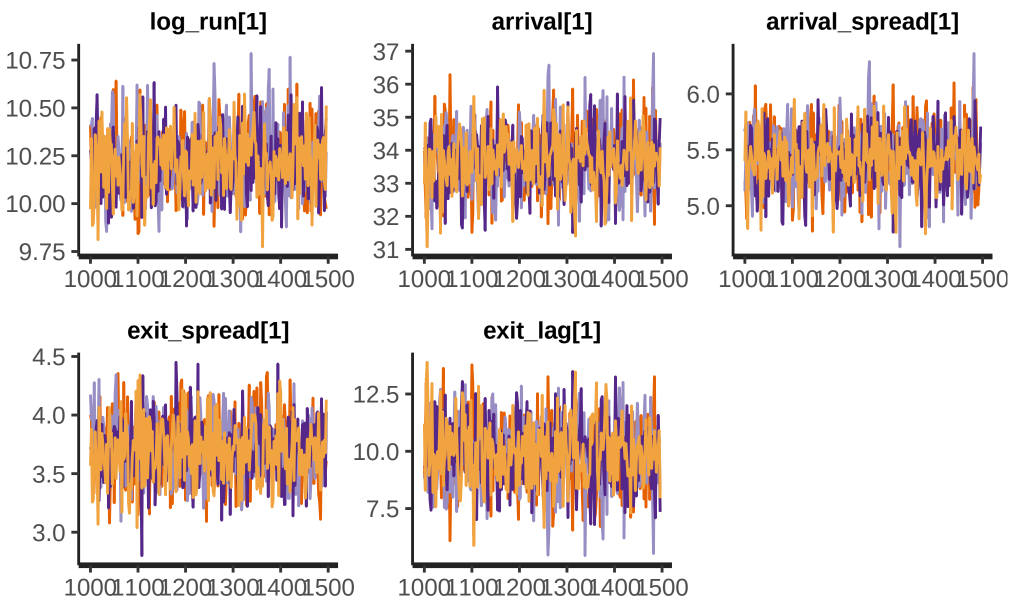


Table 2

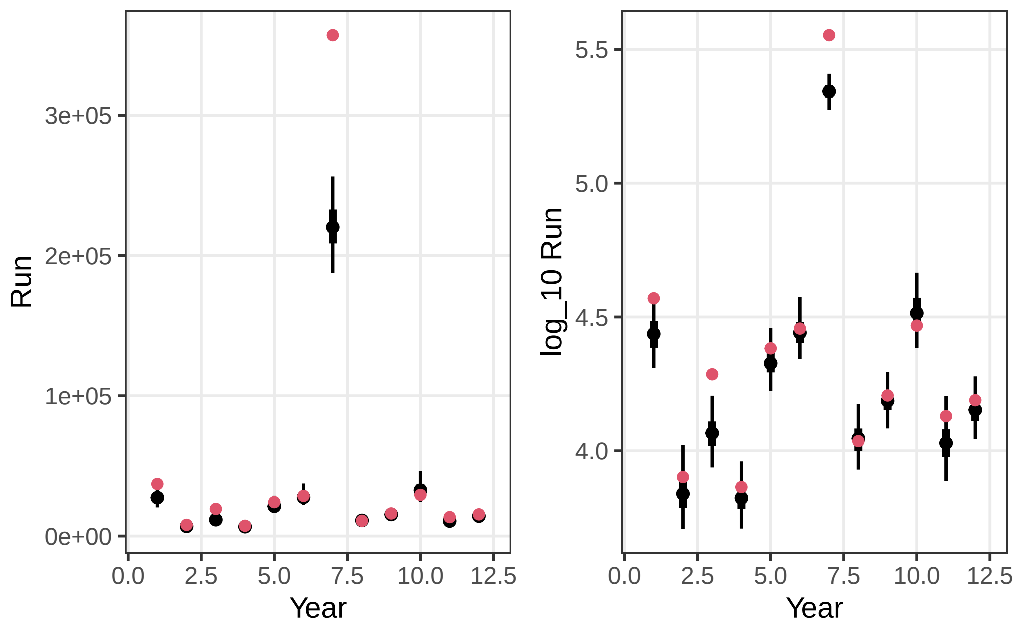
```
sp_dat = list(
    n_priors     = nrow(priors),
    priors      = data.matrix(priors[,3:4]),
    n_years     = years_n,
    n_obs       = nrow(sampled.df2),
    year        = sampled.df2$year,
    day         = sampled.df2$day,
    live_counts = sampled.df2$sampled,
    live_phi    = rnbinom_size,        # 10
    arrival_mu  = arrival_timing_mu )  # 35
```





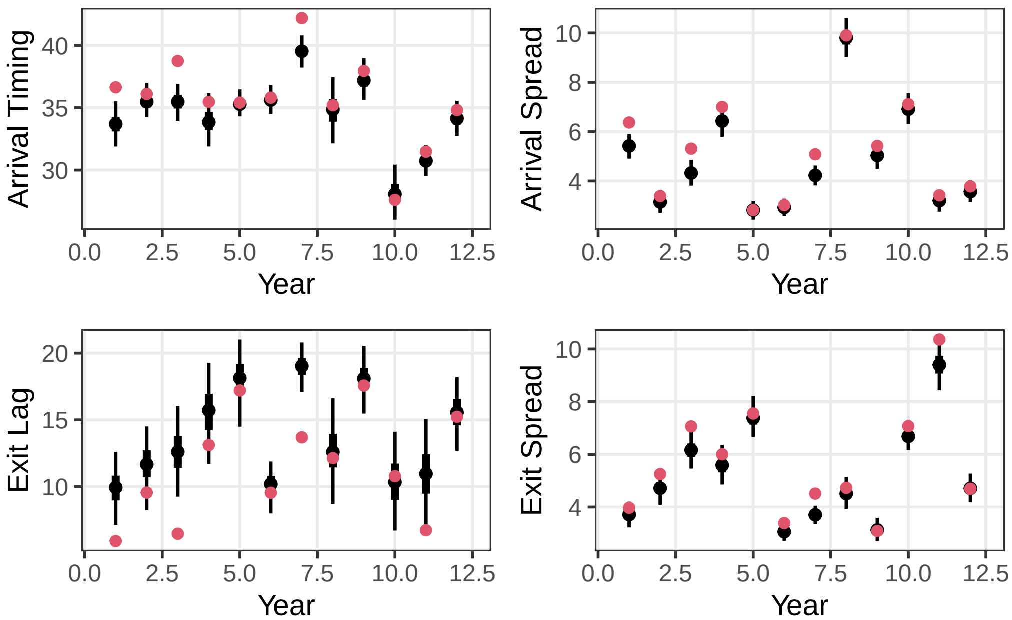
Fitted runs

Spawner abundance estimates presented logged and arith*met*ic plots. log_10 / log_e = 0.4342945



Fitted parameters

The annual values for spawner *arrival_timing*, *arrival_spread*, *exit_lag*, and *exit_spread* are presented as marginal posterior distributions (confidence limits) and compared to the simulated value. The goal is finding any consistent bias.



Spawner presence

The pattern of spawner distribution is recreated from the fitted parameters and presented with 95% confidence limits. This is compared to the observations withsampling error.

Stats

For comparison of scenarios of survey quality. Uses median absolute difference (MAD) from log (observed) - log (expected). It is faster to compute log (observed/expected). The many samples representing the posterior PDD for observed are reduced to median. mad(x) expands the value by 1.4826 to be the same as 1 stdev if x is a normal distribution. A mad of 0.16 is an accuracy of16%, implying 95%CL is $\pm 2 \ \times \ \text{MAD} \ = 68\%$ to 132%.

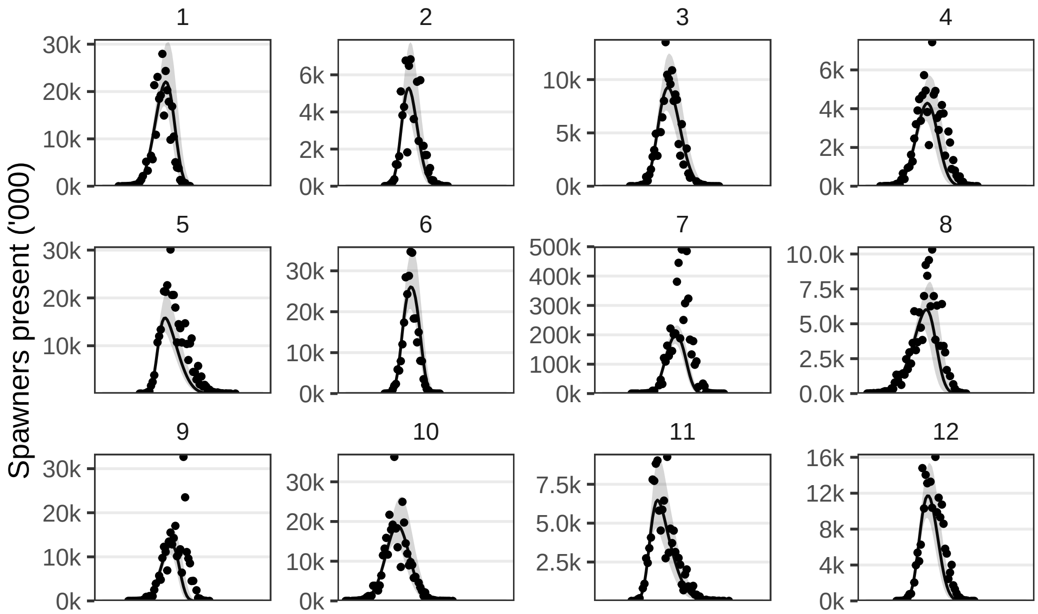


Figure 5: Estimated spawner presence, with 95% CL, from fitted parameters. Observed values are points.

Table 3: Statistics of annual parameters as ratios, each year's fitted median divided by the annual draw for that parameters before sampling error. Values for all years are summarized as median, to show bias, and mad (median absolute difference) to show precision.

Parameters	Median	MAD	CL95
Run	89%	12%	75% to 125%
Arrival Timing	98%	2%	95% to 105%
Arrival Spread	93%	6%	89% to 111%
Exit Lag	113%	16%	68% to 132%
Exit Spread	93%	5%	90% to 110%